

Fluid Flow Through Fluidized Beds

Final Lab Report
Unit Operations Lab 1
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1. Executive Summary

This experiment aimed to describe the fluidization properties of gas-solid systems and the minimum fluidization velocity of a particulate bed, which is very crucial in designing and operation of industrial fluidized bed reactors and contactors. The use of fluidized beds in chemical and energy-processing industries is extensive because of the good heat and mass transfer properties that the fluidized beds have, and the prediction of pressure drop, and fluidization is important to ensure the reliable functioning of the process.

A vertical plexiglass fluidized bed column was used in carrying out the experiment whereby, two separate experimental runs were performed, one using silica sand particles and the other using ion-exchange resin particles. Rotameters were used to control airflow, and a manometer was used to measure the pressure drop across the bed. At the two starting heights of the bed (6 in and 10 in), experiments were conducted. Data on pressure drops were recorded relative to the increasing and decreasing velocity of superficial gas in order to capture the effects of fluidization and hysteresis. Sieve analysis was used to determine the particle size distributions to aid in theoretical calculations. In the packed-bed regime, the pressure drop across the bed escalated because of the gas velocity and peaked at the start of the fluidization. After this stage, the pressure drop did not change much even when the gas velocity increased further. The minimum fluidization velocities were experimentally measured by the two types of particles and compared to those determined by the existing correlations. Under the minimum fluidization velocity, obtained pressure drops were in reasonable agreement with Ergun equation predictions. The findings indicate that particle characteristics and bed height have great effects on fluidization characteristics and minimum fluidization velocity. Although theoretical correlations were very good in giving qualitative agreement, deviations could be predicted by non-ideal particle shape, size distribution, and experimental uncertainties. (ChE 382 Unit Operations Laboratory, 2023).

2. Introduction (10 pts)

The objective of this experiment was to study the fluidization characteristics of gas solid systems through the measurement of pressure drop across particulate beds in relation to superficial gas velocity and determining the minimum fluidization velocity. These parameters play a key role in the design, scale-up, and operation of industrial fluidized bed systems, where the proper prediction of pressure drop and transitions between packed-bed and fluidized regime is directly linked to the efficiency of the processes, their safety of operation as well as product quality (McCabe, Smith, and Harriott, 2005).

The advantages of fluidized beds are that they have a good heat and mass transfer property, even temperature profile, and can be used with solid reactants or catalysts, this is the reason why these are utilized in chemical and energy-processing industries a lot. Important industrial uses are fluid catalytic cracking in petroleum refining, combustion of coal and biomass, gasification, polymerization reactors, drying and coating (Kunii and

Levenspiel, 1992; Perry and Green, 1997). Poor fluidization of such systems may lead to either channeling, high pressure drops, entrainment of the particles or unstable operation of the reactor, and importantly that experimental validation of theoretically based fluidization models in practical environments.

There was a total of 4 experiments that were carried out. The two particulate systems that were individually tested were silica sand and ion-exchange resin beads. In every particle system, each of the two bed heights of 6 inches and 10 inches were used. In every experiment, the superficial gas velocity was raised at a slow rate to a point exceeding minimum fluidization and lowered to a zero point again. This caused the experiment to have two measurement runs (increasing and decreasing velocity), which enabled the hysteresis effects to be noticed and measured.

The experimental system was a 4-inch plexiglass vertical fluidized bed column with a porous metal distributor plate to support the particulate bed and provide uniform distribution of gas. At the bottom of the column, air was injected, the flow rates were controlled by control valves and measured by rotameters. A manometer was used to measure the pressure drop across the bed during every run.

Some of the parameters that will be considered in this study would be superficial gas velocity, pressure drop, particle diameter, and minimum fluidization velocity. The comparison of experimental results with the predictions of the Ergun equation of packed-bed flow and the correlations of minimum fluidization velocity was made to determine their limitations in real non-ideal particulate systems in industrial practice.

3. Theory (10 pts)

Fluidized beds exhibit a distinct transition in hydrodynamic behavior as the superficial gas velocity is increased from values characteristic of fixed packed beds to conditions where solid particles are fully suspended and mobile. At low velocities, the fluid flows through the void spaces of a stationary bed, and the pressure drop increases with increasing velocity. As the velocity approaches a critical value, the drag force exerted by the fluid becomes sufficient to balance the effective weight of the particles, resulting in the onset of fluidization. Beyond this point, further increases in gas velocity do not produce a significant increase in pressure drop, as additional flow passes through the bed in the form of bubbles rather than through interstitial voids (ChE 382 Unit Operations Laboratory, 2023).

The relationship between pressure drop and superficial gas velocity for a fluidized bed is illustrated schematically in Figure 1 of the Lab Manuel:

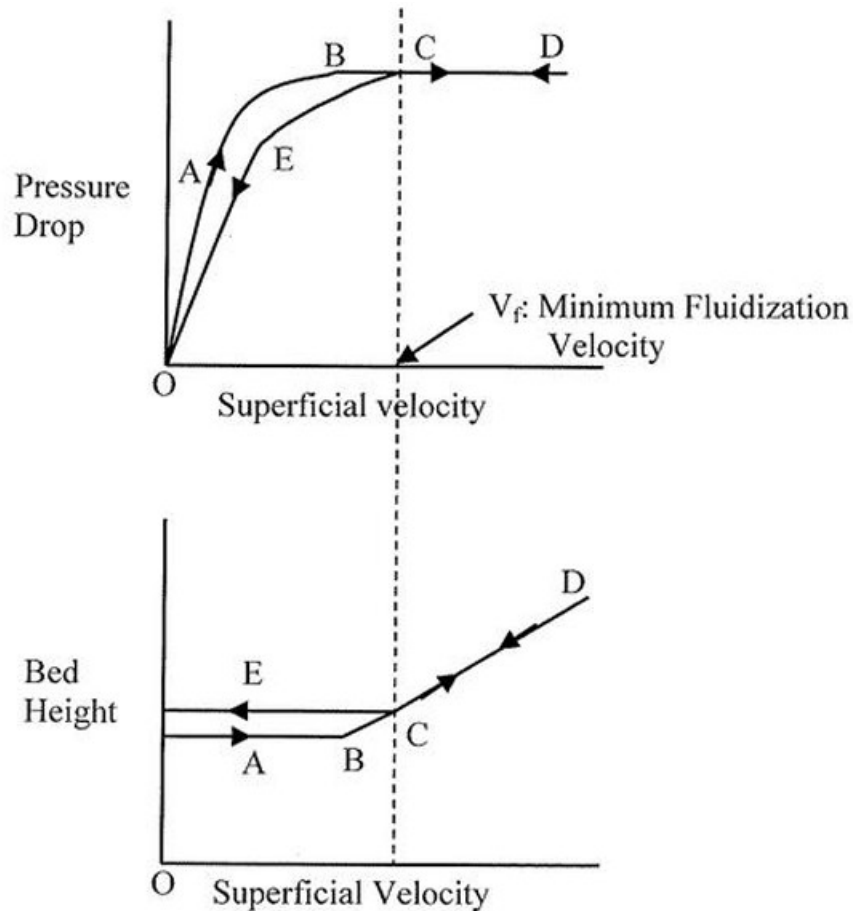


Figure 1: Pressure Drop and bed porosity or superficial air velocity.

The above figure shows both increasing and decreasing flow paths. At low velocities, the system behaves as a packed bed, and pressure drop rises steeply with velocity. At the minimum fluidization velocity, the pressure drop reaches a maximum and then plateaus. When gas velocity is subsequently decreased, the pressure drop follows a different path due to hysteresis effects arising from interparticle friction and bed restructuring (ChE 382 Unit Operations Laboratory, 2023; Kunii & Levenspiel, 1991).

In the packed-bed regime, the pressure drop per unit bed height is well described by the Ergun equation, which accounts for both viscous and inertial contributions to flow resistance through a bed of particles. For spherical or near-spherical particles, the Ergun equation is:

$$\frac{\Delta P}{L} = \frac{150 \bar{V}_0 \mu (1 - \epsilon)^2}{\Phi_s^2 D_p^2 \epsilon^3} + \frac{1.75 \rho \bar{V}_0^2 (1 - \epsilon)}{\Phi_s D_p \epsilon^3}$$

(McCabe et al., 2005, pp 160-165)

Where ΔP is the pressure drop across the bed, L is the bed height, μ is the fluid velocity, ρ is the fluid density, $\bar{V}_0$ is the superficial gas velocity, ϵ is the bed voidage, D_p

is the particle diameter, and Φ_s is the particle sphericity (ChE 382 Unit Operations Laboratory, 2023). The Ergun equation is derived from combining viscous (Darcy) and inertial (Forchheimer) contributions to fluid flow through porous media. Kunii and Levenspiel further discuss how deviations from ideal behavior can occur in real beds due to particle size distribution and non-sphericity (Kunii & Levenspiel, 1991).

In the experiment, pressure drop measurements were obtained at gas velocities below the minimum fluidization velocity and were compared directly to values predicted by the Ergun equation to assess its applicability to the experimental system.

The minimum fluidization velocity, $\bar{V}_{0M}$, represents the condition at which the upward drag force exerted by the fluid equals the effective weight of the particle bed. It's at this point that the bed begins to expand and individual particles become mobile. Because of the complex flow patterns and particle interactions in fluidized systems, the minimum fluidization velocity is commonly estimated using semi-empirical correlations based on force balances and dimensionless groups such as the particle Reynolds number (ChE 382 Unit Operations Laboratory, 2023).

McCabe, Smith, and Harriott present a correlation for minimum fluidization velocity derived from balancing pressure drop with the gravitational force on the bed, as discussed in Chapter 7 of Unit Operation of Chemical Engineering (McCabe et al., 2005, pp. 177-189). This correlation requires knowledge of particle size, particle density, fluid properties, and bed porosity at minimum fluidization. In this experiment, particle diameters were determined via sieve analysis, and literature or measured values of bed porosity were used to calculate theoretical values of the minimum fluidization velocity.

One widely used empirical correlation for minimum fluidization velocity was developed by Wen and Yu (1966), in which they related the Reynolds number at minimum fluidization to the Archimedes number:

$$Re_{mf} = [33.7^2 + 0.0408 Ar]^{1/2} - 33.7, \text{ Where } Ar = \frac{D_p^3 \rho_f (\rho_s - \rho_f) g}{\mu^2}$$

The correlation provides a practical method for estimating the minimum fluidization velocity and is frequently used in industrial fluidized-bed design (Wen & Yu, 1966).

Experimentally, the minimum fluidization velocity was determined from plots of pressure drop per unit bed height versus superficial gas velocity. The transition from the rising pressure-drop region to the plateau region identifies the minimum fluidization velocity. Experimental values of the minimum fluidization velocity obtained for sand and resin at bed heights of 6 inches and 10 inches were compared to predictions from the McCabe and Smith correlation. Agreement between experimental and theoretical values provided validation of the underlying fluidization theory, while deviations were attributed to particle non-sphericity, size distribution effects, and uncertainties in bed porosity (ChE 382 Unit Operations Laboratory 2023).

Beyond minimum fluidization, the bed expanded and gas passed through in the form of bubbles. The hydrodynamics of this regime have been outlined by Kunii and Levenspiel (1991) and Davidson et al. (1995), where the authors describe bubble formation, gas bypassing, and solid mixing behavior in detail. The existence of a

pressure-drop plateau in the fully fluidized regime arises because additional gas flow travels through bubbles with relatively low hydrodynamic resistance.

Additionally, the presence of hysteresis between increasing and decreasing flow curves were examined to illustrate the role of interparticle friction and bed structure in fluidized systems. Together, these analyses allowed for a comprehensive evaluation of packed-bed and fluidized-bed hydrodynamics using both empirical correlations and experimental observations.

4. Results (15 pts)

A condensed table is provided to reflect the $V_{0,m}$ values found graphically from each of the bed height and material conditions.

Table 1: Condensed values of $V_{0,m}$ for the different materials, heights, and flow rate +/-.

Material and Height	Flow Rate	$V_{0,m}$ (LPM)
Sand, 6 inches	Increase (+)	21.7
	Decrease (-)	32.5
Sand, 10 inches	Increase (+)	21.7
	Decrease (-)	25.3
Resin, 6 inches	Increase (+)	64.0
	Decrease (-)	85.4
Resin, 10 inches	Increase (+)	85.4
	Decrease (-)	85.4

Below are graphs from each of the trials. For each material at each bed height there is a graph for the pressure drop values divided by the length values collected as the flow rate increased, and as the flow rate decreased.

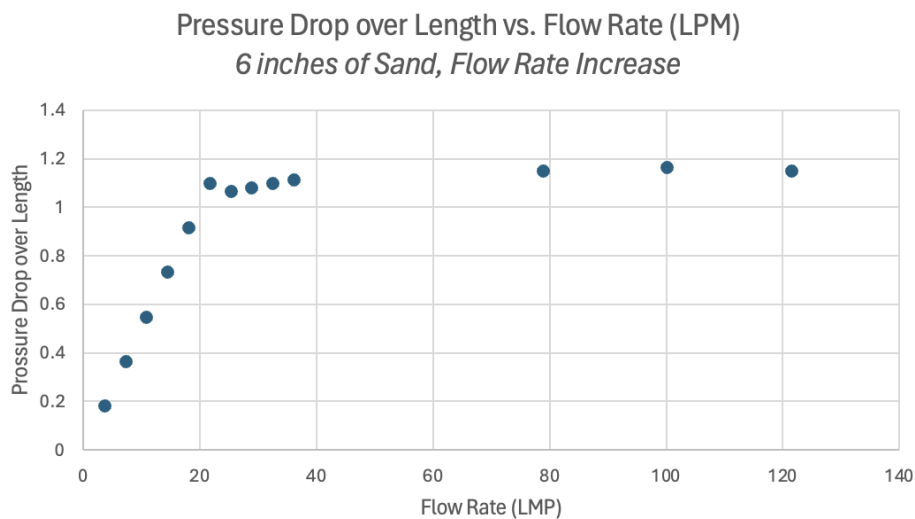


Figure 1: The response of pressure drop divided by the length as flow rate is varied, for values collected for 6 inches of sand, while the flow rate was being increased.

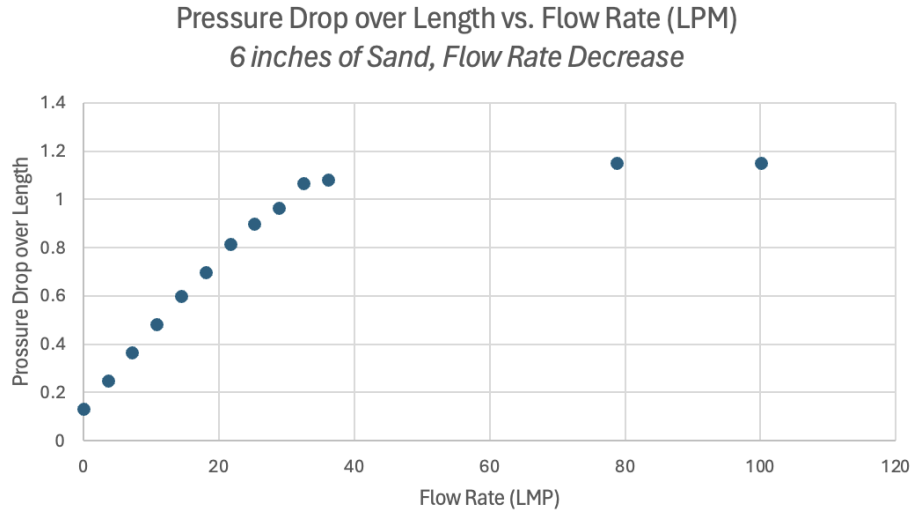


Figure 2: The response of pressure drop divided by the length as flow rate is varied, for values collected for 6 inches of sand, while the flow rate was being decreased.

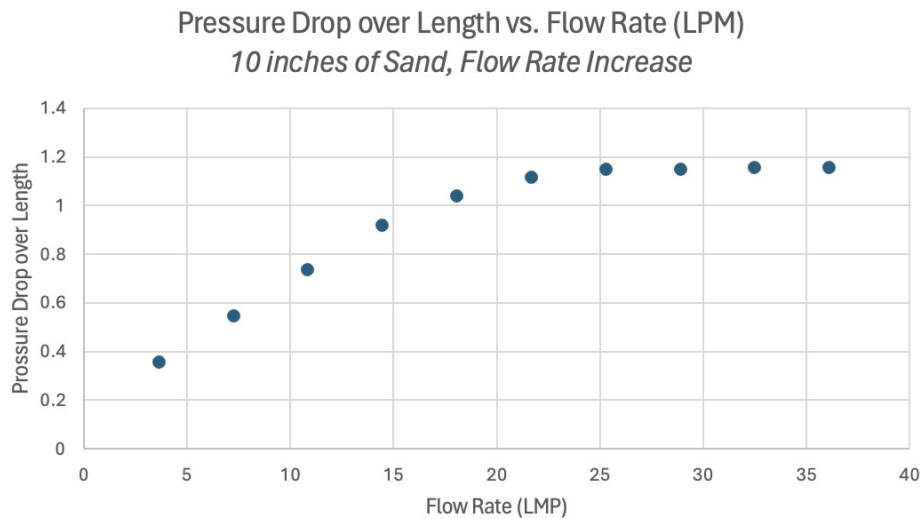


Figure 3: The response of pressure drop divided by the length as flow rate is varied, for values collected for 10 inches of sand, while the flow rate was being increased.

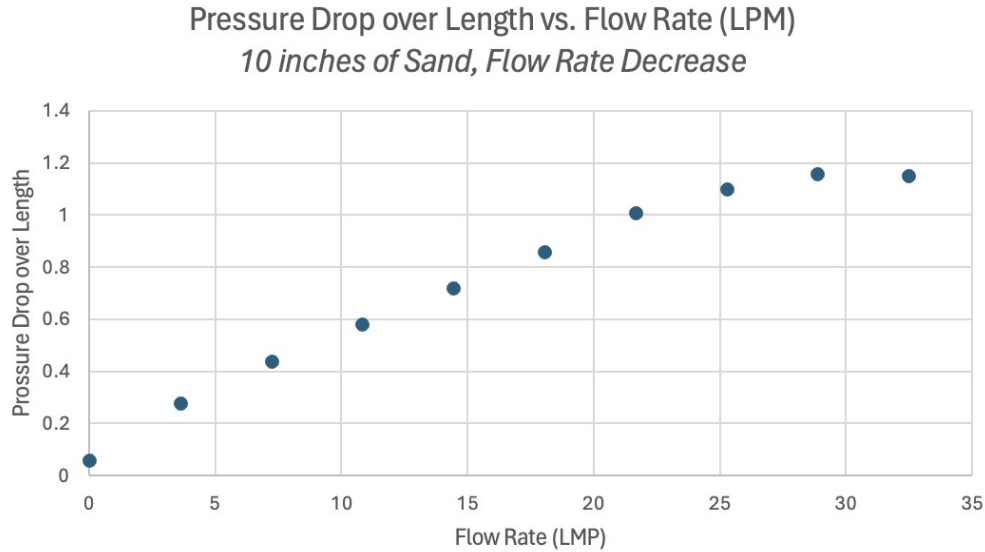


Figure 4: The response of pressure drop divided by the length as flow rate is varied, for values collected for 10 inches of sand, while the flow rate was being decreased.

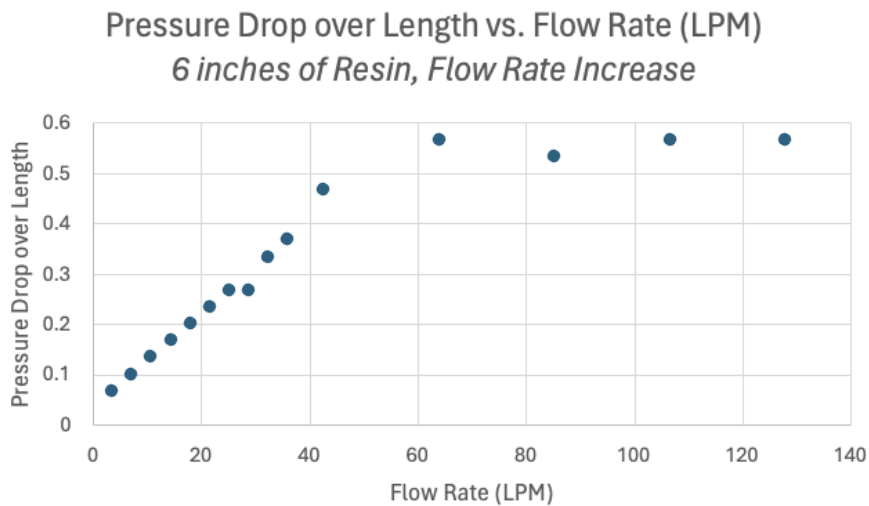


Figure 5: The response of pressure drop divided by the length as flow rate is varied, for values collected for 6 inches of resin, while the flow rate was being increased.

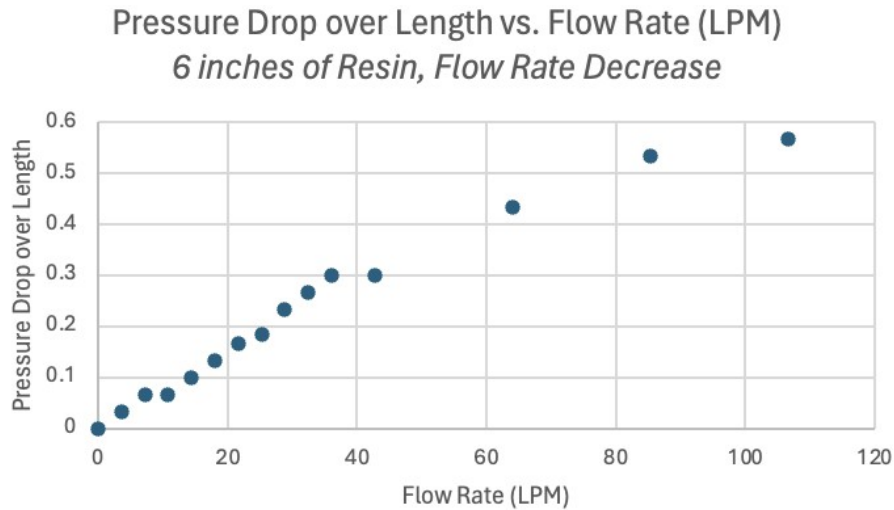


Figure 6: The response of pressure drop divided by the length as flow rate is varied, for values collected for 6 inches of resin, while the flow rate was being decreased.

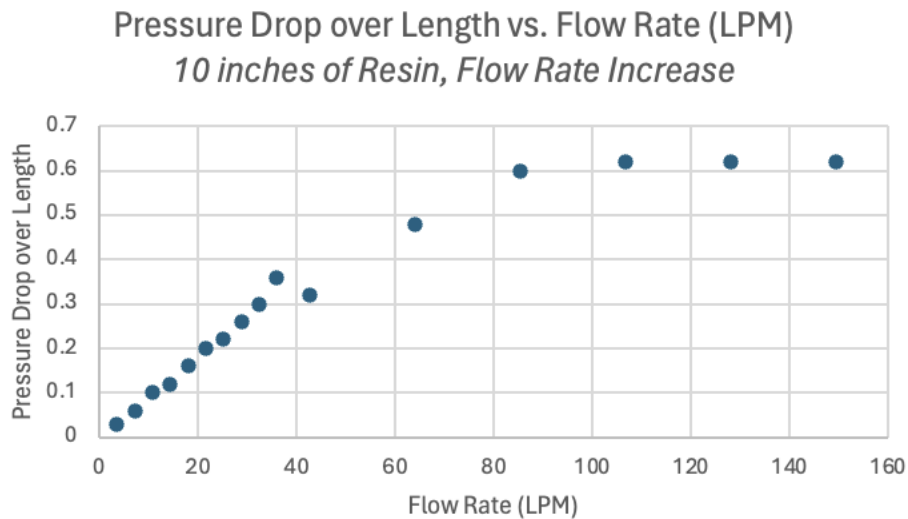


Figure 7: The response of pressure drop divided by the length as flow rate is varied, for values collected for 10 inches of resin, while the flow rate was being increased.

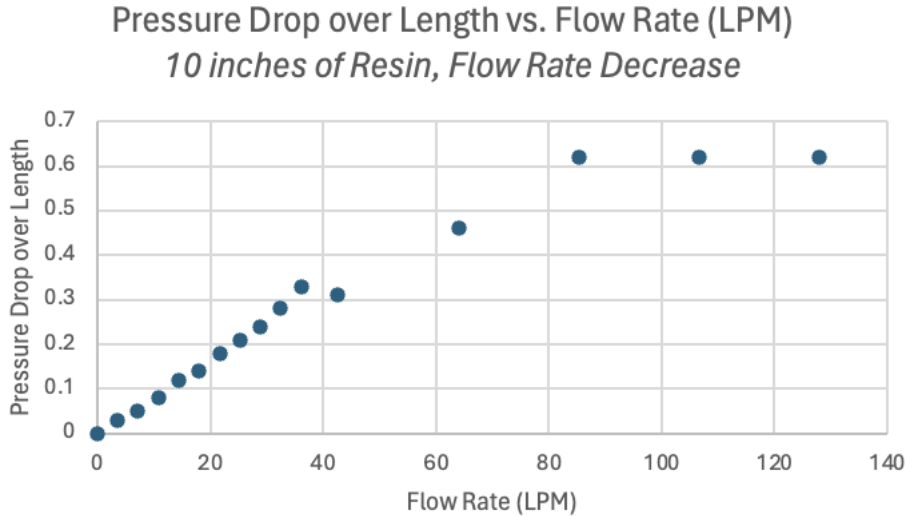


Figure 8: The response of pressure drop divided by the length as flow rate is varied, for values collected for 10 inches of resin, while the flow rate was being increased.

5. Discussion (15 pts)

The pressure drop behavior observed for both the 6-inch and 10-inch sand beds follows the characteristic trends predicted by classical gas–solid fluidization theory. At low superficial gas velocities, the pressure drop increased approximately linearly with increasing flow rate, indicating fixed-bed behavior in which flow resistance is governed by viscous and inertial losses through the packed particles. This regime is well described by the Ergun equation and is expected when particles remain stationary (Kunii and Levenspiel, 1991). As the gas velocity increased further, the pressure drop reached a plateau, indicating the onset of minimum fluidization. At this point, the upward drag force exerted by the gas balances the effective weight of the bed particles, and additional increases in velocity no longer result in a significant increase in pressure drop (Kunii and Levenspiel, 1991).

The ion-exchange resin displayed fluidization behavior consistent with expectations based on its measured particle size distribution, which was concentrated mostly in the 420–595 μm range with very few smaller particles, indicating a relatively narrow PSD compared to sand. For the 6-inch bed, the minimum fluidization velocity was estimated at approximately 0.05 m/s, comparable to sand when expressed as superficial velocity, but the pressure drop at fluidization was substantially lower (1.4–2.2 in vs. 6.6–7.0 in for sand). The lower pressure drop reflects the resin's reduced particle density, narrow PSD, and spherical geometry, which reduce the drag force required for fluidization (Helfferich, 1995; Kunii and Levenspiel, 1991). The pressure-drop plateau was smoother and more gradual than that observed for sand, consistent with literature reports that beds composed of narrowly sized, polymeric particles exhibit more stable and uniform fluidization (LeVan et al., 1997). Minor hysteresis was observed during defluidization, which is attributed to particle rearrangement and interparticle friction and is consistent with prior studies of polymeric ion-exchange beads (Kunii and Levenspiel, 1991). These results highlight the importance of particle density, measured size distribution, and shape in

determining fluidization behavior, and they provide a useful comparison to the more heterogeneous and heavier sand bed studied under identical conditions.

The experimentally determined minimum fluidization velocity (U_{mf}) was obtained by converting the measured volumetric flow rates to superficial velocity using the cross-sectional area of the 4-inch diameter column. Based on the onset of the pressure-drop plateau, U_{mf} was estimated to be approximately 0.04–0.06 m/s. Literature values for silica sand particles in the approximate size range of 200–600 μm typically report minimum fluidization velocities between 0.03 and 0.07 m/s under ambient air conditions, depending on particle size, density, and gas properties (Wen and Yu, 1966; Kunii and Levenspiel, 1991). The close agreement in magnitude indicates that the experimental results are consistent with both established theory and prior experimental work.

As expected from theory, the pressure drop at minimum fluidization increased with bed height. The 10-inch bed exhibited a significantly larger plateau pressure drop than the 6-inch bed due to the increased mass of particles that must be supported by the gas flow. However, the superficial velocity at which fluidization occurred was approximately the same for both bed heights, consistent with the theoretical prediction that U_{mf} depends primarily on particle and fluid properties rather than bed height (Kunii and Levenspiel, 1991). This agreement further supports the validity of the experimental observations.

The sand used in this experiment exhibited a broad particle size distribution, ranging approximately from 125 to 595 μm . Literature studies indicate that beds with wide particle size distributions tend to display smoother transitions to fluidization and less sharply defined pressure-drop plateaus compared to beds composed of narrowly sized particles (Chyang and Huang, 2012). This behavior is attributed to smaller particles filling void spaces and altering bed porosity, leading to a range of local fluidization velocities within the bed. The gradual onset of the plateau observed in the present data is consistent with this reported behavior and may explain minor deviations from idealized, sharply defined U_{mf} values predicted by correlations developed for uniform particle systems.

Some hysteresis was observed between the fluidization and de-fluidization curves, with pressure drop values during de-fluidization differing slightly from those measured during increasing flow. This behavior has been widely reported in fluidization studies and is typically attributed to particle rearrangement, bed compaction, and interparticle friction effects that are not fully reversible when flow rate is reduced (Kunii and Levenspiel, 1991). Such hysteresis is therefore considered a realistic feature of gas–solid fluidized beds rather than an experimental error.

Several limitations of the experimental method may have influenced the precision of the results. Identifying the exact onset of minimum fluidization from pressure-drop data involves some judgement, particularly when the transition occurs over a range of flow rates rather than at a single, sharply defined point. Additionally, uncertainties in flow rate measurement, pressure reading resolution, and bed packing reproducibility may have contributed to variability in the measured data. These issues could be mitigated in future

experiments by performing multiple trials, averaging U_{mf} values, using finer flowrate increments near the transition region, and improving instrumentation accuracy.

The assumptions made in this experiment such as steady-state flow, negligible particle clumping, constant gas flow, and uniform bed distribution are generally reasonable and consistent with those used in standard fluidization models. However, assumptions of spherical particles and uniform particle size are less reliable given the broad particle size distribution of the sand, which likely contributed to deviations from ideal model predictions. Despite these limitations, the assumptions were sufficient to capture the dominant physical behavior of the system. It could benefit to sieve the entire mass, not just an arbitrary 100 grams, that enters the column to get a true particle size distribution.

Definite conclusions that can be drawn from this lab include the confirmation that the pressure-drop behavior and minimum fluidization velocity of sand in a gas-solid system are consistent with classical fluidization theory, and that U_{mf} is largely independent of bed height, while the pressure drop at fluidization does change with bed weight. More speculative conclusions involve the precise quantitative agreement with empirical correlations, as these correlations are sensitive to particle size distribution and shape, which were not fully controlled in this experiment.

The results of this experiment reinforce the use of classical fluidization models to practical systems and highlight the importance of particle size distribution when interpreting experimental data. From an engineering perspective, these findings support the practice of operating fluidized beds slightly above U_{mf} to ensure stable fluidization while minimizing energy consumption.

Several questions remain unanswered and suggest future work. Further experiments could investigate the effect of using narrowly sized particles, different particle densities, or alternative fluidizing gases to better isolate the influence of individual parameters on U_{mf} . Although minor, the temperature could also play a role considering we assumed room temperature in winter.

6. References (10 pts)

ChE 382 Unit Operations Laboratory. *Fluid Flow Through Fluidized Beds Lab Manual*; Rev. Spring 2023v2; University of Illinois at Chicago: Chicago, IL, 2023.

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<https://doi.org/10.1021/ie9607307>

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7. Appendix I: Data Tabulation/Graphs (10 pts)

Data in Tables 5-9 include lab collected values, and calculated values for flow rate and pressure drop divided by length. Before is in reference to the flow rate increase, and after is in reference to the flow rate decrease. Various notes were made as data was collected for both a visual reference as the particles becoming fluidized and potential sources of error within the data collection. Table 9 is repeating the same trial completed in Table 8 to assess error.

Table 2: Values collected for resin particle size distribution.

Resin Particle Size Distribution Data			
Sieve size (µm)	Mass before (g)	Mass after (g)	Difference
595	594	670	76
500	562	570	8
420	561	574	13
354	535	537	2
250	524	524	0
210	541	541	0
125	501	501	0

Table 3: Values collected for sand particle size distribution.

Sand Particle Size Distribution Data			
Sieve size (µm)	Mass before (g)	Mass after (g)	Difference
595	594	594	0.1
500	562	563	0.2
420	561	561	0.0
354	354	354	0.0
250	524	554	30.1
210	541	551	10.8
125	501	559	58.5

Table 4: Values collected for void fraction calculations.

Void Fraction Data		
Material	Solid Volume (mL)	Water Volume Used (mL)
Resin	17	11
Sand	11	6

Table 5: Values collected for 6 inches of sand.

Sand 6 in. Bed Height						
Fluidization				"De-fluidization"		
Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)		Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)
4	1.1	0.18		100	6.9	1.15
7	2.2	0.37		79	6.9	1.15
11	3.3	0.55		36	6.5	1.08
14	4.4	0.73		32	6.4	1.07
18	5.5	0.92		29	5.8	0.97
22	6.6	1.10		25	5.4	0.90
25	6.4	1.07		22	4.9	0.82
29	6.5	1.08		18	4.2	0.70
32	6.6	1.10		14	3.6	0.60
36	6.7	1.12		11	2.9	0.48
79	6.9	1.15		7	2.2	0.37
100	7.0	1.17		4	1.5	0.25
121	6.9	1.15		0	0.8	0.13

Table 6: Values collected and calculated for 10 inches of sand.

Sand 10 in. Bed Height						
Fluidization				"De-fluidization"		
Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)		Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)
4	3.6	0.36		32	11.5	1.15
7	5.5	0.55		29	11.6	1.16
11	7.4	0.74		25	11	1.1
14	9.2	0.92		22	10.1	1.01
18	10.4	1.04		18	8.6	0.86
22	11.2	1.12		14	7.2	0.72
25	11.5	1.15		11	5.8	0.58
29	11.5	1.15		7	4.4	0.44
32	11.6	1.16		4	2.8	0.28
36	11.6	1.16		0	0.6	0.06

Table 7: Values collected for 6 inches of resin.

Resin 6 in. Bed Height						
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Fluidization			"De-fluidization"		
Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)	Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)
4	0.4	0.07	107	3.4	0.57
7	0.6	0.1	85	3.2	0.53
11	0.8	0.13	64	2.6	0.43
14	1	0.17	43	1.8	0.30
18	1.2	0.2	36	1.8	0.30
22	1.4	0.23	32	1.6	0.27
25	1.6	0.27	29	1.4	0.23
29	1.6	0.27	25	1.1	0.18
32	2	0.33	22	1	0.17
36	2.2	0.37	18	0.8	0.13
43	2.8	0.47	14	0.6	0.10
64	3.4	0.57	11	0.4	0.07
85	3.2	0.53	7	0.4	0.07
107	3.4	0.57	4	0.2	0.03
128	3.4	0.57	0	0	0.00

Table 8: Values collected for 10 inches of resin.

Resin 10 in. Bed Height					
Fluidization			"De-fluidization"		
Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)	Flow Rate (LPM)	ΔP (in)	$\Delta P/L$ (in/in)
4	0.3	0.03	128	6.2	0.62
7	0.6	0.06	107	6.2	0.62
11	1	0.1	85	6.2	0.62
14	1.2	0.12	64	4.6	0.46
18	1.6	0.16	43	3.1	0.31
22	2	0.2	36	3.3	0.33
25	2.2	0.22	32	2.8	0.28
29	2.6	0.26	29	2.4	0.24
32	3	0.3	25	2.1	0.21
36	3.6	0.36	22	1.8	0.18
43	3.2	0.32	18	1.4	0.14
64	4.8	0.48	14	1.2	0.12
85	6	0.6	11	0.8	0.08
107	6.2	0.62	7	0.5	0.05
128	6.2	0.62	4	0.3	0.03
149	6.2	0.62	0	0	0

Table 9: Values collected and calculated for 10 inches of resin – a repeated trial for error analysis.

8. Appendix II: Error Analysis (10 pts)

In this experiment, particle size distributions were determined along with void fraction, pressure drop, and minimum fluidization velocities for sand and resin beds at two initial bed heights of 6 in and 10 in. Systemic, random, and human errors affected the calculated particle size, void fraction, pressure drop, and experimentally determined minimum fluidization velocity.

The average particle diameters calculated were 548.94 μm for resin and 155.68 μm for sand. Sources for uncertainty in sieve analysis included balance precision (± 0.1 g assumed), loss of fine particles during transfer, and incomplete separation between sieve sizes. Given ~ 100 g total sample mass, a ± 0.1 balance uncertainty gave a fractional uncertainty of $0.1/100 = 0.1\%$. However, the main error likely came from particle size distribution assumptions when calculating the average diameter. The harmonic averaging method amplifies small errors in fine particle fractions, particularly for sand where most mass was in the 125-250 μm range. The estimated uncertainty in average particle diameter was $\pm 5\text{-}8\%$ for sand and $\pm 3\text{-}5\%$ for resin.

From the water displacement data, the resin void fraction was ~ 0.65 and for sand was ~ 0.50 . The graduated cylinder precision was assumed to be ± 1 mL and bed volume difference was $\sim 10\text{-}15$ mL, the relative uncertainty was $1/10 = 10\%$. Void fraction had a strong effect on theoretical minimum fluidization velocity. The estimated void fraction uncertainty was $\pm 8\text{-}12\%$. This likely contributed significantly to differences between theoretical and experimental fluidization velocities.

Pressure drop was read from a manometer in inches of water. Resolution was approximately ± 0.1 in water. For sand in the 6 in bed, $\Delta P(\text{plateau}) \approx 1.15$ in. The relative uncertainty was $0.1/1.15 \approx 8.7\%$. For resin in the 6 in bed, $\Delta P(\text{plateau}) \approx 0.62$ in and the relative uncertainty was $0.1/0.62 \approx 16\%$. Thus, the pressure drop uncertainty was much larger for resin, which explained the greater variability in the experimental minimum fluidization velocity.

The theoretical fluidization velocity for sand was determined to be $\approx 4.35\text{-}4.92$ LPM and the experimental value was calculated to be ≈ 4.35 LPM (6"). This was determined to be a very good agreement, within 10%. Percent error for sand (6") was determined to be 11.6%:

$$\% \text{ error} = \frac{|4.35 - 4.92|}{4.92} \times 100 \approx 11.6\%$$

For resin, the theoretical fluidization velocity was determined to be $\approx 11.8\text{-}12.3$ LPM and the experimental value was calculated as ≈ 12.3 LPM with an agreement within

~5%. However, the resin pressure drop plateau was less sharply defined and showed instability. The overall uncertainty in the minimum fluidization velocity for sand was ± 10 -15% and ± 15 -20% for resin.

A rotameter was estimated to have a $\pm 5\%$ accuracy. At low flow (3-10 LPM), this introduced $\approx \pm 0.2$ -0.5 LPM and since the minimum fluidization velocity was determined visually at a transition region, this contributed to uncertainty.

During the experiment, vigorous bubbling, rotameter fluctuations, and trembling were observed which suggested non-uniform fluidization, channeling, distributor non-ideal flow, and gas bypassing. These behaviors affected the resin runs in particular, where bubbling began gradually and the plateau was less clearly defined. This introduced subjective uncertainty in identifying the exact onset of fluidization.

Static bed heights of 6 in and 10 in were assumed uniform. Small errors (± 0.25 in) in bed height affect $\Delta P/L$ and since L appears in the denominator, it affected calculated pressure gradient. Estimated height uncertainty was $\pm 4\%$.

To assess repeatability of the fluidization measurements, the 10 in resin bed was run twice under the same nominal conditions. The two trials produced very similar pressure-drop plateaus and identified the onset of fluidization in essentially the same flow region, indicating that the experiment had reasonable run-to-run precision for resin despite the qualitative instability observed at higher flows (i.e. vigorous bubbling).

Despite measurement uncertainty, both sand and resin exhibited clear pressure-drop plateaus. Experimental minimum fluidization velocity values agreed with theoretical predictions with 5-15% and expected trends were observed. Therefore, the experimental uncertainty did not prevent valid conclusions. The agreement between experiment and theory was within expected laboratory accuracy. To reduce errors in future experiments, using a digital differential pressure sensor instead of a manual manometer would give more accurate results. Performing multiple replicate runs to calculate statistical confidence intervals. Also, performing more flow increments near the transition region would better define the minimum fluidization velocity.

9. Appendix III: Sample Calculations (10 pts)

The average particle diameter calculations for sand are done by taking the mass difference for each sieve from before and after the sand was sifted, summing the differences, and then dividing the individual mass differences of each sieve by the sum of the differences to get the mass fraction. The individual mass fractions are then divided by the sieve size, summed, and the inverse value of the sum is taken. For example, the 420 micron sieve weighed 560.9 grams before the sand was sifted, and 574 grams after the sand was sifted. The weight difference is 13.1 grams. The sum of the differences across all of the sieves was 98.3 grams. 13.1grams divided by 98.3 grams is 13.32%, or a 0.1332 mass fraction. The mass fraction is divided by the sieve size, 420 micron, to get 0.00317 microns. The sum of the mass fractions divided by their respective sieve size and having the inverse taken is 548.9 microns.

The void fraction for sand is calculated using the difference in volume of water poured to wet the sand, divided by the volume of sand being wetted. The void fraction is equal to $(48-37)/17$, which is 0.65.

The following calculations are for values found at 6 inches of sand, once the pressure stabilized, which is the 11th reading of the flow rate increase in **Table**. During the lab, the pressure drop values were calculated while the lab was being performed, as to verify when pressure stabilized. The difference in inches found when the pressure drop stabilized was 6.9 in. Divided by the 6 inches of sand height for the column – dP/L was found to be 1.15.

To solve for the flow rate, the percent value read from the manometer was multiplied by the manometer flow rate value. The flow rates of the two different manometers are added. This is calculated as $1 \times 36.09 + 0.1 \times 426.8 = 78.77$ LPM.

Using equation 7.51 from *Unit Operations of Chemical Engineering* by McCabe et al., and excels solver, the theoretical value of V_f is solved for. The sphericity used came from the values within McCabe et al., as 0.8. A random guess is made for $V_{0,m}$ or V_f within a cell in excel and used in equation 7.51. Solver is used to set the value found for equation 7.51 to 1.15, as that is the dP/L value at stabilization, and the cell containing V_f should be used to change in order to get the values to match. The theoretical value of V_f is found to be 4.35 LPM.

10. Appendix IV: Individual Team Contributions

NAME: Kendall

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	
Pre-Lab Editing	1	Color coded and labeled apparatus, listed materials and their purpose.
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0.5	Color coded and numbered apparatus, created a table to explain numbering and color code.
Materials & Supplies (prelab)	0.25	Created a table for the materials and supplies and their purpose
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0.25	Did the safety chart
Final Lab Editing	1	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	

Results (final lab)	4	Read the McCabe textbook to learn more about sphericity and equation 7.51, completed the calculations, made the tables and charts.
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	2	Set up the data tables and organized them, created the graphs from the data
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	1	Detailed how I completed the calculations found in the results section.
Power Point Presentation	1	Transferred information from the pre-lab to the power point
Total	13	

NAME: Lindsey

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Spent 1 hour during prelab to determine the particle diameter using sieves and another hour on lab day to read the pressure drop during the experiment.
Pre-Lab Editing	1	Outlined the experimental procedure.
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	1	Used the lab manual to determine a step by step procedure.
Project Safety Evaluation (prelab)	0	
Final Lab Editing	1	Theory and error analysis formatting
Executive Summary (final)	0	

lab)		
Introduction (Final lab)	0	
Theory (Final Lab)	2	Used provided sources as well as outside sources to outline the theory behind the experiment.
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0.5	Added references to the report.
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	2	Made simple calculations to determine errors, outline uncertainty, sources of error, and how to prevent errors in the future
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Transferred information from the report to the presentation
Total	10.5	

NAME: Christian

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Conducted Pre-lab setups
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	.5	Evaluated safety of the lab experiment
Final Lab Editing	1	Checked over other sections for logic
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		

Discussion (final lab)	2	Found literature and compared to our findings
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	Took time to ensure our data was presented neatly
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	Helped in the safety and procedure sections
Total	8.5	

NAME: Srija

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Helped determine the experimental objective and the experimental and helped with the measurement from the control valves
Pre-Lab Editing	1	
Experimental Objective (Prelab)	2	Helped the determine the objective
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	1	Indroduction and executive summary formatting
Executive Summary (final lab)	2	Did the executive summary
Introduction (Final lab)	2	Did do the indtroduction
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	1	Used references and added in the section
Data Tabulation/Graphs (final)	0	

Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Transferred information from the pre lab to the presentation
Total	12	